Fiscal Year:	FY 2025	**Task Last Updated:**	FY 07/09/2025
PI Name:	Carter, Dorothy Ph.D.		
Project Title:	Project FUSION: Facilitating Unified Systems of Interdependent Organizational Networks		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	48824	**Congressional District:**	7
Comments:	The Principal Investigator (PI) was previously at the University of Georgia.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-Crew Health (FLAGSHIP, OMNIBUS). Appendix A-Omnibus, Appendix B-Flagship
Start Date:	02/12/2018	**End Date:**	08/21/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Per the Principal Investigator (PI), the following researchers are no longer Co-Investigators with this project: Noshir Contractor, Ph.D.; Aaron Schecter, Ph.D.; Leslie DeChurch, Ph.D. (Ed., 7/31/25).		
COI Name (Institution):	Shuffler, Marissa L (Clemson University)		
Grant/Contract No.:	80NSSC23K1124 ; 80NSSC18K0511		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As we set our sights on Mars, and other destinations beyond lower Earth orbit, we must enable extreme forms of teamwork across Spaceflight Multiteam Systems (SFMTSs) composed of teams that are separated by unprecedented degrees of space and time. In “Project FUSION: Facilitating Unified Systems of Interdependent Organizational Networks,” we are engaging in a transformative research program rooted in the past decade of theory and research on MTSs, but breaking new ground in how MTSs are conceptualized and studied. Our programmatic research aims to illuminate the underlying forces that give rise to the psycho-social relational states (e.g., influence, trust, shared cognition) within and between teams that underpin mission success. These crucial relationships, and the drivers of their emergence, will need to be understood, monitored, and at times, circumvented using countermeasures in order to enable coordinated efforts across the MTSs involved in Long-duration Exploration Mission (LDEMs). Project FUSION is a multi-pronged, multi-method, interdisciplinary project with three main research foci: (1) field investigations with NASA personnel; (2) development of an agent-based computational model of spaceflight MTS functioning; and (3) controlled laboratory experiments and analog studies. Our research design is intended to be iterative. Findings within each foci are continually infusing the refinement and design of research in other foci. Project FUSION is an applied research project with the ultimate goal of translating findings from three research foci in order to provide NASA with a "countermeasure toolkit" comprised of validated interventions that can be used to facilitate effective teamwork in spaceflight MTSs. The countermeasure toolkit consists of: (1) a MTS task analysis procedure, (2) code and recommendations based on our agent-based computational model of spaceflight MTS dynamics, (3) a multiteam training countermeasure ready for operational implementation with astronauts and mission controllers, and (4) a multiteam debriefing protocol ready for operational implementation with astronauts and mission controllers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The findings from this project could have substantial implications for human life on Earth, and in particular, for the effectiveness of teams and larger systems of teams in organizations operating in high-stakes environments. The field studies, laboratory studies, and computer simulation studies aim to better understand the patterns of social relationships (e.g., collective attention, trust, influence, information sharing) that are likely to form within and across teams in large interdependent organizational systems. By better understanding the patterns of relationships that are likely, we can help determine when and where teamwork interventions or "countermeasures" are necessary. Moreover, the ultimate goal of this applied research project is to develop and validate a toolkit of countermeasures, including training, debriefing procedures, and decision-making protocols that are designed to facilitate team and inter-team collaboration in complex organizational systems. These countermeasures will be able to be utilized widely across many organizational contexts beyond NASA (e.g., healthcare, the military, corporations).
Task Progress:	As NASA prepares for Long-duration Exploration Missions (LDEMs) such as a future mission to Mars, the challenges of supporting extreme teamwork across Spaceflight Multiteam Systems (MTSs) are becoming more urgent and complex. Project FUSION (“Facilitating Unified Systems of Interdependent Organizational Networks”) is a multi-method, interdisciplinary research initiative designed to understand and improve teamwork across the multiple interdependent teams involved in space missions—such as astronauts, mission controllers, engineers, and medical staff. The central aim of this project is to uncover the key drivers of effective coordination and communication across these teams and to provide NASA with actionable tools (“countermeasures”) that can mitigate risks and improve performance during future missions. Over the course of the project, we have made significant progress across all three core research tracks: field investigations, laboratory experiments, and computational modeling. In our fieldwork, we conducted in-depth interviews with over 30 former and current NASA mission control personnel. These interviews revealed deep insights into the relational dynamics—such as influence, trust, and attention—that enable (or hinder) coordination across the larger system of mission teams. In parallel, we conducted detailed archival analysis of over 60 years of mission logs and historical documentation, helping us trace the evolution of multiteam coordination practices over time and across mission contexts. We also partnered with hospital systems to examine how MTS coordination can break down in real-world high-stakes environments on Earth and how communication patterns and processes can support MTS performance. These observational/interview studies further validated our core insights and helped us tailor intervention strategies for use in both space and terrestrial mission contexts. In addition to field studies, we completed a series of controlled experiments within three of NASA’s Human Exploration Research Analog (HERA) campaigns and leveraged these human subjects' data to develop agent-based computational models of spaceflight MTS functioning. These studies simulated the conditions of future Mars missions—such as communication delays and crew autonomy—to examine their impact on multiteam functioning. Our results consistently indicate that communication delays disrupt collective attention across teams, especially when combined with elevated autonomy. These findings emphasize the importance of supporting crew coordination through interventions like training for shared leadership, promoting message clarity, and increasing awareness of inter-team dependencies. Findings from all three research foci are being translated into a suite of countermeasures, including: • A structured task analysis procedure to identify MTS risks, • A computational model of SFMTS dynamics with accompanying recommendations for relational and communication dynamics, • A multiteam training protocol to strengthen shared mental models and leadership capacity, and • A multiteam debriefing procedure designed to enhance team learning and alignment across mission stages. Project FUSION continues to operate as an iterative research initiative, with each line of inquiry informing the others. Our next steps include testing and validating the full countermeasure toolkit and ensuring it is operationally viable for NASA training and mission readiness programs.
Bibliography Type:	Description: (Last Updated: 08/05/2025)
Abstracts for Journals and Proceedings	Bauer L, Carter DR, Weinberger C, Bausman M, Shuffler M, Landon L. "Leading autonomous component teams in multiteam systems: Insights from NASA." 2025 Society for Industrial and Organizational Psychology (SIOP) Annual Conference, Denver, Colorado, April 2-5, 2025. Abstracts. 2025 Society for Industrial and Organizational Psychology (SIOP) Annual Conference, Denver, Colorado, April 2-5, 2025. , Apr-2025

Abstracts for Journals and Proceedings	Pearman J, Carter DR, Lungeanu A, Bauer L, Carter N, Pendergraft J, Landon L, Shuffler M, DeChurch L, Contractor N. "Collective attention in virtual teams: Mitigating the negative impact of communication delays on team performance." 2025 Society for Industrial and Organizational Psychology (SIOP) Annual Conference, Denver, Colorado, April 2-5, 2025. Abstracts. 2025 Society for Industrial and Organizational Psychology (SIOP) Annual Conference, Denver, Colorado, April 2-5, 2025. , Apr-2025
Abstracts for Journals and Proceedings	Carter DR, Lungeanu A, Pearman J, Carter N, Bauer L, Pendergraft J, Landon L, Shuffler M, DeChurch L, Contractor N. "Collective attention in spaceflight: Mitigating the negative impact of communication delays on multiteam system performance." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Shuffler ML, Carter DR, Bauer L, Weinberger C, Pearman J, Gerkin E, Bausman M, Landon L. "From Earth to Mars: NASA Mission Control insights on challenges and adaptations of future spaceflight missions." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Books/Book Chapters	Bauer L, Weinberger C, Carter DR, Landon LB. "Managing spaceflight team stress: Considerations for multiteam system research." in "Stress and Well-Being in Teams." Ed. P. Harms, C-H. Chang. Leeds, UK: Emerald Publishing Limited, 2024. p. 171., Dec-2024